

Numerical Methods in Option Pricing: Monte Carlo and Variance Reduction Techniques

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Abstract

The Monte Carlo method is a numerical technique for estimating expectations via repeated random sampling and is widely used in quantitative finance for derivative pricing when closed-form solutions are unavailable or computationally expensive.

This study develops a Monte Carlo framework under the Geometric Brownian Motion model and benchmarks European option prices against the Black–Scholes analytical solution. Variance reduction techniques, specifically antithetic, and control variates, are implemented and their impact on convergence and estimator variance is analysed across increasing simulation sizes. The framework is further extended to estimate option sensitivities (Greeks) through a bump-and-revalue approach and to accommodate path-dependent derivatives such as Asian options and to derive the P&L identity of a continuously delta-hedged option position.

Results demonstrate that variance reduction methods materially improve convergence rates and reduce variance relative to standard Monte Carlo simulation, while remaining consistent with Black–Scholes benchmarks.

1 Introduction

Pricing derivatives is crucial in quantitative finance as help keep a transparent framework for Over-The-Counter deals, ensures no-arbitrage consistency under the model assumptions, they also allow risk management such as delta hedging in order to protect a desk against adverse market movements[1].

The Monte Carlo method is used to approximate expectations under a risk neutral measure. It is particularly effective for path dependent options, where closed form solutions are . However, it suffers from slow convergence $O\left(\frac{1}{\sqrt{N}}\right)$, which motivates the use of variance reduction techniques.

This report implements Monte Carlo pricing under Geometric Brownian Motion whilst benchmarking against Black-Scholes closed form solutions for European options, the investigation of antithetic variates and control variates against standard Monte Carlo. The framework also includes computation of Greeks via bump and revalue and the path dependent Asian options.

The study focuses on quantifying the efficiency gains from variance reduction techniques and analysing their impact on convergence behaviour.

2 Theoretical Framework

2.1 Risk-Neutral Valuation

Throughout this paper, we assume an arbitrage-free market. By the First Fundamental Theorem of Asset Pricing, there exists at least one equivalent martingale (risk-neutral) measure, denoted by $\mathbb{Q}$ [2]. Under this measure, all discounted asset prices are martingales. Consequently, the value of a contingent claim with payoff X at maturity T is given by

$$V_0 = e^{-rT} \mathbb{E}_{\mathbb{Q}}[X][3],$$

which forms the basis for the Monte Carlo valuation approach used in this study[3].

The significance of the risk-neutral measure is that it allows derivative prices to be expressed as discounted expectations of future payoffs. Rather than requiring an estimate of investors' risk preferences or expected market returns, valuation can be performed solely using the risk-free rate and the probability distribution of the underlying asset under $\mathbb{Q}$.

For a European call option with strike K and underlying asset price S_T at maturity, the payoff is

$$X = (S_T - K)^+.$$

The price of the option is therefore

$$C_0 = e^{-rT} \mathbb{E}_{\mathbb{Q}}[(S_T - K)^+]. \quad (1)$$

The remaining challenge is the computation of the expectation

$$\mathbb{E}_{\mathbb{Q}}[(S_T - K)^+].$$

Furthermore, the GBM assumption is consistent with important qualitative properties of option prices. Since the payoff of a European call option,

$$(S_T - K)^+,$$

is a convex function of the underlying asset price, Jensen's inequality implies that the corresponding discounted option price process is a submartingale under the risk-neutral measure. Intuitively, uncertainty increases the expected value of a convex payoff, reflecting the fact that option holders benefit from favourable price movements while their downside remains limited. In this study, this expectation is estimated using Monte Carlo simulation under the assumption that the underlying asset follows a geometric Brownian motion.[2]

2.2 Geometric Brownian Motion

To evaluate the expectation appearing in the risk-neutral pricing formula, a model for the dynamics of the underlying asset is required. Throughout this study, the underlying asset price is assumed to follow a Geometric Brownian Motion (GBM). Under the risk-neutral measure $\mathbb{Q}$, the asset drift is equal to the risk-free rate, i.e. $\mu = r$, and the asset pricing

$$dS_t = \mu S_t dt + \sigma S_t dW_t[3],$$

where μ denotes the risk-free interest rate, σ is the volatility of the asset, and W_t is a standard Brownian motion under $\mathbb{Q}$.

The GBM model is widely used in quantitative finance due to its analytical tractability and its consistency with several stylised features of financial markets. In particular, it implies that asset prices remain strictly positive and that continuously compounded returns are normally distributed. These properties lead directly to the Black-Scholes pricing framework and allow the asset price at time T to be expressed explicitly as

$$S_T = S_0 \exp \left[\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z \right], \quad (2)$$

where $Z \sim N(0, 1)$.

This result can be derived using Itô's Lemma[4]. Suppose the stock price follows the stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t. \quad (3)$$

Let

$$f(S_t) = \ln(S_t).$$

Applying Itô's Lemma,

$$df(S_t) = f'(S_t) dS_t + \frac{1}{2} f''(S_t) (dS_t)^2. \quad (4)$$

Since

$$f'(S_t) = \frac{1}{S_t}, \quad f''(S_t) = -\frac{1}{S_t^2},$$

it follows that

$$d(\ln S_t) = \frac{1}{S_t} dS_t - \frac{1}{2 S_t^2} (dS_t)^2. \quad (5)$$

Squaring Equation (3) gives

$$(dS_t)^2 = \mu^2 S_t^2 (dt)^2 + \sigma^2 S_t^2 (dW_t)^2 + 2\mu\sigma S_t^2 dt dW_t.$$

Using Itô's multiplication rules,

$$(dW_t)^2 = dt, \quad (dt)^2 = 0, \quad dt dW_t = 0,$$

we obtain

$$(dS_t)^2 = \sigma^2 S_t^2 dt. \quad (6)$$

Substituting dS_t and $(dS_t)^2$ into Equation (5) yields

$$d(\ln S_t) = \frac{1}{S_t} (\mu S_t dt + \sigma S_t dW_t) - \frac{1}{2 S_t^2} (\sigma^2 S_t^2 dt),$$

which simplifies to

$$d(\ln S_t) = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t. \quad (7)$$

Integrating from 0 to T ,

$$\ln(S_T) - \ln(S_0) = \left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma W_T. \quad (8)$$

Exponentiating both sides gives

$$S_T = S_0 \exp \left[\left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma W_T \right]. \quad (9)$$

Since

$$W_T \sim N(0, T),$$

it can be written as

$$W_T = \sqrt{T}Z, \quad Z \sim N(0, 1).$$

Hence,

$$S_T = S_0 \exp \left[\left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z \right]. \quad (10)$$

Under the risk-neutral measure introduced in the previous section, $\mu = r$, giving the terminal stock price distribution used throughout the Monte Carlo simulations:

$$S_T = S_0 \exp \left[\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z \right]. \quad (11)$$

[6] This closed-form representation is particularly convenient for Monte Carlo simulation, as it allows asset prices to be generated directly from independent samples of a standard normal random variable.

An important consequence of this final expression is that S_T is log-normally distributed under an equivalent martingale measure, ensuring that the stock price is strictly positive whilst compounded returns are normally distributed. The sum of the properties discussed in this section is what makes the Geometric Brownian Motion model so widely used in derivative pricing.

Although empirical asset returns exhibit features not captured by GBM, such as stochastic volatility, jumps and volatility clustering, the model provides an effective benchmark for studying numerical pricing methods. Its analytical tractability allows Monte Carlo estimators to be validated against known Black-Scholes solutions before extending the framework to more complex path-dependent derivatives.

2.3 Black–Scholes Model

The Black-Scholes model provides a closed-form solution for the price of European options under the assumptions introduced in the previous section. Its analytical tractability makes it a natural benchmark against which Monte Carlo estimators can be validated. The model assumes frictionless markets, continuous trading, constant volatility, a constant risk-free interest rate, the absence of arbitrage opportunities, no dividend payments, and an underlying asset whose dynamics follow a geometric Brownian motion. The Black-Scholes price of a European call option is given by

$$C = S_0 N(d_1) - K e^{-rT} N(d_2),$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T},$$

and $N(\cdot)$ denotes the cumulative distribution function of a standard normal random variable. Here, S_0 is the current asset price, K the strike price, r the risk-free interest rate, σ the volatility of the underlying asset, and T the time to maturity. The formula highlights the importance of volatility in option pricing. Since the payoff of a call option is convex in the underlying asset price, greater uncertainty increases the expected payoff under the risk-neutral measure, leading to higher option values. The formula also admits an intuitive interpretation. Differentiating the option price with respect to the underlying asset price yields

$$\frac{\partial C}{\partial S_0} = N(d_1)$$

hence

$$\Delta = N(d_1).$$

The quantity $N(d_1)$ therefore represents the hedge ratio of the option, that is, the number of shares required to locally replicate or delta-hedge the position. Furthermore, $N(d_2)$ can be interpreted as the risk-neutral probability that the option expires in-the-money, since

$$\mathbb{Q}(S_T > K) = N(d_2)$$

while Ke^{-rT} represents the present value of the strike price payable at maturity[5]. Taken together, these components provide an intuitive interpretation of the Black-Scholes formula: the option price is the cost of constructing a self-financing replicating portfolio whose terminal payoff exactly matches that of the option. The Black-Scholes model plays two important roles throughout this study. First, it provides the theoretical foundation for risk-neutral valuation and delta hedging. Second, its closed-form solution allows Monte Carlo estimators to be validated against an exact benchmark. Any discrepancy between the Monte Carlo estimate and the Black-Scholes price can therefore be attributed to simulation error rather than model error. As the number of simulated paths increases, the Monte Carlo estimator is expected to converge towards the Black-Scholes value. [5]

2.4 Monte Carlo Pricing

2.4.1 Monte Carlo Estimator

The Monte Carlo simulation was first proposed for option pricing by Boyle [7], it uses repeated random sampling to model probability and predict outcomes under uncertainty relying on the law of large numbers for the convergence of the average of random paths towards the outcome. The risk-neutral valuation formula derived in Section 2.1 requires the evaluation of the expectation of the option payoff under the risk-neutral measure. For many path-dependent derivatives, this expectation cannot be computed analytically, making Monte Carlo simulation a natural numerical alternative. The method approximates the expectation by repeated random sampling.

From the terminal stock price distribution derived in Equation (10), independent samples

$$Z_1, \dots, Z_N \sim N(0, 1)$$

are generated. Each sample produces one simulated terminal asset price,

$$S_T^{(i)} = S_0 \exp \left[\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z_i \right],$$

from which the corresponding European call payoff is obtained as

$$X_i = (S_T^{(i)} - K)^+.$$

The Monte Carlo estimator of the option price is therefore

$$\hat{C}_N = e^{-rT} \frac{1}{N} \sum_{i=1}^N X_i.$$

This estimator is unbiased since,

$$\mathbb{E}[\hat{C}_N] = C_0.$$

Since the simulated payoffs are independent and identically distributed, the Law of Large Numbers implies that the estimator converges almost surely to the true option price as the number of simulations tends to infinity [8],

$$\hat{C}_N \xrightarrow[N \rightarrow \infty]{a.s.} C_0.$$

Consequently, increasing the number of simulated paths improves the accuracy of the estimator, although this convergence occurs at the relatively slow rate of $O(N^{-1/2})$, motivating the variance reduction techniques developed in the following sections[8].

2.4.2 Statistical Error and Convergence

To compare the efficiency of different Monte Carlo estimators, it is necessary to quantify the statistical uncertainty associated with the estimated option price. The variance of the option price estimator is given by

$$Var(\hat{C}_N) = Var(e^{-rT} \frac{1}{N} \sum_{i=1}^N X_i)$$

which simplifies to,

$$Var(\hat{C}_N) = e^{-2rT} Var(\frac{1}{N} \sum_{i=1}^N X_i)$$

and since the samples are independent

$$Var(\sum_{i=1}^N X_i) = \sum_{i=1}^N Var(X_i) = N\sigma_X^2$$

where $\sigma_X^2 = Var(X)$. Hence,

$$Var(\hat{C}_N) = e^{-2rT} \frac{\sigma_X^2}{N}$$

taking the square root gives

$$SE(\hat{C}_N) = e^{-rT} \frac{\sigma_X}{\sqrt{N}}$$

Since σ_X is generally unknown, it is estimated by the sample standard deviation

$$s_X = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X})^2}$$

which gives the estimator used in the rest of this report,

$$SE(\hat{C}_N) = e^{-rT} \frac{s_X}{\sqrt{N}} [8],$$

which is reported alongside every simulated option price.

This confirms the earlier claim that the convergence is of the order of $O(\frac{1}{\sqrt{N}})$. Furthermore by the Central Limit Theorem,

$$\sqrt{N}(\hat{C}_N - C_0) \xrightarrow{d} N(0, \sigma_X^2),$$

By the Central Limit Theorem, the Monte Carlo estimator is asymptotically normally distributed[8]. Consequently, the standard error provides a natural measure of the estimator's statistical uncertainty and is therefore reported alongside every simulated option price throughout this report.

As previously stated,

$$SE(\hat{C}_N) = O\left(\frac{1}{\sqrt{N}}\right)$$

This highlights the slow convergence of standard Monte Carlo. Since the standard error decreases proportionally to $N^{-\frac{1}{2}}$, halving the estimation error requires approximately four times as many simulation paths. Additionally, each new simulation requires the generation of one or more random numbers (depending on whether the payoff is path dependent), the simulation of the asset paths, and the evaluation of the payoff. Each additional simulation substantially increases the computational cost. Since the Monte Carlo estimator is itself random, repeated executions of the algorithm produce slightly different option prices. As the number of simulated paths increases, this variability decreases according to the $N^{-\frac{1}{2}}$ convergence rate. Consequently, improving Monte Carlo efficiency is generally achieved by reducing the variance of the estimator rather than simply increasing the number of simulations. The following chapter investigates two classical variance reduction techniques: antithetic variates and control variates.

3 Variance Reduction

3.1 Introduction

Since the computational cost of Monte Carlo simulation increases linearly with the number of simulated paths, improving efficiency is generally achieved by reducing the variance of the estimator rather than simply increasing the sample size.

3.2 Antithetic variate

3.2.1 Motivation

Standard Monte Carlo simulation generates one payoff from each random sample Z . Since the standard normal distribution is symmetric, the corresponding sample $-Z$ is equally valid and can be used to generate a second payoff at essentially no additional random number generation cost. Averaging the two payoffs reduces random fluctuations because they are negatively correlated, thereby reducing the variance of the estimator.

3.2.2 Construction of the estimator

Let,

$$X^+ = X(Z),$$

and

$$X^- = X(-Z)$$

If

$$Z \sim N(0, 1),$$

then

$$-Z \sim N(0, 1).$$

Consequently, both $S_T(Z)$ and $S_T(-Z)$ are valid terminal asset prices, hence defining the antithetic payoff as,

$$X_{AV} = \frac{1}{2}(X^+ + X^-),$$

where

$$X^+ = (S_T(Z) - K)^+.$$

Therefore defining the antithetic estimator as,

$$\hat{C}_{AV} = e^{-rT} \frac{1}{N} \sum_{i=1}^N X_{AV}^{(i)}.$$

This estimator is implemented by pairing each simulated standard normal random variable with its negation, allowing two terminal asset prices to be generated from every random draw. [8]

3.2.3 Unbiasedness

To establish unbiasedness, it suffices to show that the expected value of the estimator equals the true option price. Since both Z and $-Z$ are distributed by a standard normal,

$$\mathbb{E}[X^+] = \mathbb{E}[X^-].$$

Therefore,

$$\mathbb{E}[X_{AV}] = \frac{1}{2}(\mathbb{E}[X^+] + \mathbb{E}[X^-]) = \mathbb{E}[X^+].$$

Hence,

$$\mathbb{E}(\hat{C}_{AV}) = C_0.$$

3.2.4 Variance Analysis

The variance of the antithetic estimator can be expressed as,

$$Var(X_{AV}) = Var\left(\frac{X^+ + X^-}{2}\right) = \frac{1}{4}Var(X^+ + X^-).$$

Hence,

$$Var(X_{AV}) = \frac{1}{4}(Var(X^+) + Var(X^-) + 2Cov(X^+, X^-))$$

However notice that $Var(X^+) = Var(X^-)$ since,

$$X^+ \stackrel{d}{=} X^-$$

so we can substitute

$$Var(X^+) = Var(X^-) = \sigma^2.$$

By definition of the moment of correlation coefficient,

$$\rho = \frac{Cov(X^+, X^-)}{\sigma_{X^+} \sigma_{X^-}}$$

Since

$$\sigma_{X^+} = \sigma_{X^-} = \sigma,$$

it follows that

$$\begin{aligned} Cov(X^+, X^-) &= \rho \sigma^2 \\ Cov(X^+, X^-) &= \rho \sigma^2. \end{aligned}$$

Resulting in the following expression,

$$Var(X_{AV}) = \frac{1}{4}(\sigma^2 + \sigma^2 + 2\rho\sigma^2) = \frac{\sigma^2}{2}(1 + \rho).$$

It immediately follows that the variance of the antithetic estimator is lower than the standard estimator if and only if $\rho < 0$. Since the payoff of a European call option is a monotone increasing function of the terminal stock price, and the terminal stock price is itself a monotone increasing function of Z , the random variables X^+ and $X(-Z)$ are negatively correlated.

This shows mathematically that this estimator has a lower variance then the standard Monte Carlo. In practice, antithetic variates can substantially reduce estimator variance while requiring only one additional payoff evaluation per random sample.

[8][9]

3.3 Control Variates

3.3.1 Motivation

Unlike antithetic variates, which reduce variance by exploiting symmetry in the sampling process, the control variate method reduces variance by incorporating information from a related random variable whose expected value is known analytically. The key idea is to replace part of the Monte Carlo estimator with a quantity whose average is already known exactly. If the control variable is highly correlated with the target payoff, deviations in one variable are mirrored by deviations in the other, allowing a suitable correction to substantially reduce estimator variance while preserving unbiasedness.

3.3.2 Control Variate Estimator

Let,

$$\mathbb{E}[Y] = \mu_Y$$

where Y is a random control variable highly correlated with X . The estimator is constructed by correcting the original payoff using deviations of the control variable from its known expectation,

$$X_{CV} = X - \beta(Y - \mu_Y).$$

Hence,

$$\hat{C}_{CV} = e^{-rT} \frac{1}{N} \sum_{i=1}^N X_{CV}^{(i)}.$$

This is the standard control variate estimator that will be discussed in the rest of this section. In the implementation presented in this report, the control variate estimator is combined with antithetic variates by averaging paired simulations prior to applying the control correction. Since both techniques are unbiased, they may be combined without altering the expected value of the estimator.

[8][10]

3.3.3 Unbiasedness

The unbiasedness follows directly from linearity of expectation,

$$\mathbb{E}[X_{CV}] = \mathbb{E}[X] - \beta(\mathbb{E}[Y] - \mu_Y) = \mathbb{E}[X].$$

Therefore the estimator is unbiased.

3.3.4 Finding Optimal Coefficient

The reader may notice the presence of the coefficient β , which determines the magnitude of the correction applied to the original estimator. To determine the coefficient that minimises the estimator variance, consider the variance of the desired random variable,

$$Var(X_{CV}) = Var(X - \beta(Y - \mu_Y)) = Var(X - \beta Y).$$

Expanding,

$$Var(X_{CV}) = Var(X) + \beta^2 Var(Y) - 2\beta Cov(X, Y).$$

Differentiating with respect to β ,

$$\frac{d}{d\beta} Var(X_{CV}) = 2\beta Var(Y) - 2Cov(X, Y).$$

The optimal coefficient occurs when the derivative is null; therefore,

$$\beta^* = \frac{Cov(X, Y)}{Var(Y)}.$$

This choice minimises the variance of the estimator and is therefore referred to as the optimal control variate coefficient. This is the exact same coefficient calculation that's implemented in the program. Therefore, introducing the control variate changes the variance of the estimator but not its expectation.

3.3.5 Choice Of Control Variable

As stated an effective control variate should satisfy the following two conditions, its expected value must be known analytically; it should be strongly correlated with the payoff being estimated. For a European call option, a natural choice of control variable is the terminal stock price,

$$Y = S_T.$$

Since,

$$\mathbb{E}[S_T] = S_0 e^{rt},$$

and because

$$(S_T - K)^+$$

is an increasing function of S_T , the two variables are therefore strongly correlated. Therefore S_T and its expected value can be calculated analytically, so fits the requirements for the method. Hence,

$$X_{CV} = X - \beta(S_T - S_0 e^{rt}).$$

For arithmetic Asian options, a different control variable is employed. Specifically, the payoff of the corresponding geometric Asian option is used because its expected value admits a closed-form solution while remaining highly correlated with the arithmetic payoff. This construction is presented in Section 4.4.

3.3.6 Why It Works

As previously stated,

$$Var(X_{CV}) = Var(X) + \beta^2 Var(Y) - 2\beta Cov(X, Y).$$

Then we can substitute the expression for β^* as well as,

$$\rho = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}.$$

Consequently,

$$Var(X_{CV}) = Var(X) + \left(\frac{Cov(X, Y)}{Var(Y)} \right)^2 Var(Y) - 2 \frac{Cov(X, Y)}{Var(Y)} Cov(X, Y).$$

Thus,

$$Var(X) + \frac{(Cov(X, Y))^2}{Var(Y)} - 2 \frac{(Cov(X, Y))^2}{Var(Y)} = Var(X) \left(1 - \left(\frac{Cov(X, Y)}{\sigma_X^2 \sigma_Y^2} \right)^2 \right)$$

Consequently,

$$Var(X_{CV}) = Var(X)(1 - \rho^2)$$

Consequently, the effectiveness of the control variate method is entirely dependent on the correlation between the payoff and the chosen control variable. When $\rho = 0$, no variance reduction is achieved. As the magnitude of the correlation approaches one, the variance decreases according to the factor $(1 - \rho^2)$, becoming arbitrarily small when the variables are almost perfectly correlated. This result shows that the efficiency of the control variate method depends entirely on the correlation coefficient between the payoff and the chosen control variable. Since

$$0 \leq \rho^2 \leq 1,$$

the variance reduction factor is always bounded between zero and one. The closer the correlation is to unity in magnitude, the greater the reduction in estimator variance.

Notice that the sign of the correlation is irrelevant because only ρ^2 appears. Unlike antithetic variates, where negative correlation is desirable, control variates benefit equally from strong positive or negative correlation. When

$$\rho \rightarrow 1,$$

the control variable explains almost all of the variability in the original payoff. Consequently,

$$\text{Var}(X_{CV}) = \text{Var}(X)(1 - \rho^2) \rightarrow 0.$$

In the limiting case where the control variable is perfectly correlated with the payoff, the Monte Carlo estimator becomes deterministic after the correction, and additional simulations provide essentially no improvement because almost all sampling error has been eliminated. Control variates are not universally effective. Their performance deteriorates when the chosen control variable exhibits little correlation with the target payoff. If

$$\rho \approx 0,$$

then

$$\text{Var}(X_{CV}) \approx \text{Var}(X),$$

so almost no improvement is obtained. Furthermore, if the expected value

$$\mu_Y = \mathbb{E}[Y]$$

is inaccurate, the estimator may become biased. Consequently, an effective control variate must satisfy two essential conditions: its expectation must be known exactly (or with negligible error); it must be highly correlated with the quantity being estimated. These conditions motivate the choice of the terminal stock price for European options and the geometric Asian option for arithmetic Asian options.

This explains why the choice of control variable is considerably more important than the actual regression coefficient. Even a perfectly estimated coefficient cannot substantially reduce variance if the chosen control variable is only weakly correlated with the quantity of interest. In the present study, the terminal stock price is selected as the control variable for European options owing to its known expectation under the risk-neutral measure and its strong positive correlation with the option payoff. For arithmetic Asian options, the closed-form price of the corresponding geometric Asian option is employed as the control variate. The numerical effectiveness of these choices is investigated in Chapters 5 and 6. [8]

3.4 Combined Variance Reduction

3.4.1 Motivation

Since antithetic variates and control variates reduce variance through different mechanisms, they may be combined to further improve the efficiency of the Monte Carlo estimator. Because both techniques are unbiased, their combination preserves the expected value of the estimator while potentially achieving greater variance reduction than either method individually. The general idea goes as follows; for each generated random Z , use the average payoff of Z and $-Z$ and apply a control variate to that average payoff.

3.4.2 Constructing the estimator

As previously defined,

$$X_{AV} = \frac{1}{2}(X^+ + X^-).$$

Applying an antithetic variate to the control variate,

$$Y_{AV} = \frac{1}{2}(Y^+ + Y^-).$$

Finally,

$$X_{ACV} = X_{AV} - \beta(Y_{AV} - \mu_Y).$$

Hence,

$$\hat{C}_{ACV} = e^{-rT} \frac{1}{N} \sum_{i=1}^N X_{ACV}^{(i)}.$$

This estimator is still unbiased as,

$$\begin{aligned} \mathbb{E}[X_{ACV}] &= \mathbb{E}[X_{AV}] - \beta(\mathbb{E}[Y_{AV}] - \mu_Y) \\ &= \mathbb{E}[X] \end{aligned}$$

Hence, combining the two techniques preserves unbiasedness.

3.4.3 Implementation

For each simulated standard normal sample Z , an antithetic sample $-Z$ is generated. The corresponding terminal asset prices and payoffs are averaged to form the antithetic estimator. The control variate correction is then applied using the averaged control variable together with the optimal coefficient β^* estimated from the simulated sample.

3.4.4 Expected performance

Antithetic variates reduce variance by inducing negative correlation between paired simulations, whereas control variates reduce variance by exploiting the strong positive correlation between the payoff and a variable with known expectation. Since these mechanisms are fundamentally different, their effects are complementary rather than redundant. Consequently, the combined estimator is expected to achieve a lower variance than either method applied independently, although the precise improvement depends on the strength of the correlations involved.

The three estimators considered in this chapter are all unbiased estimators of the option price. Their difference lies solely in their variances. Standard Monte Carlo serves as the baseline estimator, antithetic variates reduce variance through negative correlation, while control variates exploit positive correlation with a known random variable. The combined estimator implemented in this study seeks to benefit from both mechanisms simultaneously before their numerical performance is assessed in Chapter 6.

4 Asian Option Pricing

4.1 Motivation

While European options depend solely on the terminal asset price, many financial contracts depend on the behaviour of the underlying asset throughout the life of the contract. Such derivatives are referred to as path-dependent options, since their payoff depends on the entire trajectory of the underlying asset rather than its value at maturity alone. Asian options constitute one of the most widely traded classes of path-dependent derivatives. Rather than depending exclusively on the terminal stock price, their payoff is determined by the average asset price over a predetermined observation period. Averaging reduces the impact of temporary market fluctuations and makes these contracts less susceptible to market manipulation near maturity. Compared with European options, pricing Asian options is considerably more computationally demanding. Each simulation requires the generation of an entire asset path rather than a single terminal stock price. If daily monitoring is employed over one year, a single simulation requires approximately 252 Gaussian random variables instead of one. Consequently, variance reduction becomes substantially more valuable, as reducing estimator variance often provides larger computational savings than simply increasing the number of simulated paths. This chapter first introduces arithmetic Asian options before presenting the corresponding geometric Asian option, whose closed-form solution makes it an ideal control variate. Finally, the construction of the geometric control variate estimator used throughout the numerical experiments is developed.

4.2 Arithmetic Asian Options

As previously mentioned, the payoff of an Asian option depends on the average price of the underlying asset observed at a collection of monitoring dates. Subsequently the payoff of an arithmetic Asian is given by,

$$X_A = (A - K)^+.$$

Where,

$$A = \frac{1}{M} \sum_{i=1}^M S_i$$

and M the number of times the underlying asset is monitored until maturity. Increasing the number of monitoring dates generally makes the option less sensitive to extreme movements in the underlying asset, since the payoff depends on an average rather than a single terminal observation. Unlike a European call option, whose payoff depends only on S_T , the arithmetic Asian payoff depends on every observation of the underlying asset throughout the life of the contract. Consequently, the complete asset path must be simulated when pricing the option using Monte Carlo methods. The Black-Scholes framework relies on the terminal stock price following a lognormal distribution under the geometric Brownian motion assumption. Although each individual asset price is lognormally distributed under the geometric Brownian motion assumption, the arithmetic average of lognormal random variables does not possess a lognormal distribution. Consequently, the Black-Scholes methodology cannot be applied directly, and no general closed-form pricing formula exists for arithmetic Asian options. Monte Carlo simulation provides a natural pricing framework for arithmetic Asian options. Rather than requiring an explicit probability distribution for the arithmetic average, the method estimates the risk-neutral expectation directly by simulating complete asset price paths and computing the corresponding average payoff. Each simulation generates an entire sequence of asset prices

$$S_{t_1}, S_{t_2}, \dots, S_{t_M},$$

from which the arithmetic average is computed before evaluating the discounted payoff. Repeating this procedure over many independent simulated paths yields a Monte Carlo estimate of the risk-neutral option price.

4.3 Geometric Asian Options

4.3.1 Definition

Unlike the arithmetic Asian option, whose payoff depends on the arithmetic mean of the monitored asset prices, the geometric Asian option is based on the geometric mean of the same observations. For Monitoring dates

$$t_1, t_2, \dots, t_M,$$

the geometric average is defined by

$$G = \left(\prod_{i=1}^M S_{t_i} \right)^{1/M}$$

and the corresponding call payoff is

$$X_G = (G - K)^+.$$

Although both contracts depend on the same simulated asset path, replacing the arithmetic mean by the geometric mean fundamentally changes the mathematical properties of the payoff.

4.3.2 Importance of the Geometric Average

The geometric average possesses a remarkable mathematical property that makes it analytically tractable. Unlike the arithmetic average, whose distribution is unknown under geometric Brownian motion, the geometric average remains lognormally distributed, allowing the option price to be obtained in closed form.

Taking the logarithm of the geometric average gives

$$\ln G = \ln \left(\left(\prod_{i=1}^M S_{t_i} \right)^{1/M} \right) = \frac{1}{M} \sum_{i=1}^M \ln S_{t_i}.$$

As shown in Equation (8), under geometric Brownian motion,

$$\ln S_{t_i} = \ln S_0 + \left(r - \frac{1}{2}\sigma^2 \right) t_i + \sigma W_{t_i},$$

so each log-price is normally distributed. Furthermore, since Brownian motion is a Gaussian process, the vector

$$(\ln S_{t_1}, \dots, \ln S_{t_M})$$

is jointly normally distributed. Any linear combination of jointly normal random variables is therefore also normally distributed. Since $\ln G$ is simply the arithmetic average of these log-prices, it follows that

$$\ln G \sim N(\mu_G, \sigma_G^2).$$

Exponentiating both sides gives

$$G = e^{\ln G},$$

and therefore G is itself lognormally distributed.

This property is fundamental. Whereas the arithmetic average of lognormal random variables has no tractable probability distribution and therefore admits no general closed-form pricing formula, the geometric average remains lognormally distributed. Consequently, geometric Asian options possess a closed-form pricing formula analogous to the Black-Scholes model, making them ideal control variates for pricing arithmetic Asian options.

4.3.3 Closed-form pricing

Since the geometric average remains lognormally distributed under the geometric Brownian motion assumption, the pricing problem can be reduced to the valuation of a lognormal payoff. As a result, geometric Asian options admit a closed-form solution analogous to the Black-Scholes formula. This analytical tractability is particularly useful in the present study, since the geometric Asian option will later serve as a control variate for the arithmetic Asian option.

$$C_G = e^{-rT} \mathbb{E}_Q[(G - K)^+]$$

[11] [8] [1]

4.3.4 Why it matters

Although geometric Asian options are less commonly traded than arithmetic Asian options, they play a central role in variance reduction. Their importance lies not in their standalone financial use, but in the fact that their payoff is highly correlated with the arithmetic Asian payoff while still admitting a closed-form price. This makes the geometric Asian option an ideal control variate for the arithmetic Asian option. Since both payoffs are computed from the same simulated asset path, large movements in the underlying asset typically affect both quantities in the same direction. This strong positive correlation is exactly the property required for an effective control variate. The closed-form geometric price can be computed once before the Monte Carlo simulation begins, so the control variate correction adds negligible additional computational cost relative to the simulation itself. [11] [8]

4.3.5 Comparison with arithmetic Asians

Although the arithmetic and geometric Asian options are constructed from the same monitored asset prices, their mathematical properties differ substantially. The arithmetic average of lognormal random variables does not admit a tractable distribution, which prevents the derivation of a general closed-form pricing formula. By contrast, the geometric average preserves lognormality, allowing an explicit valuation formula to be obtained. This distinction is precisely what makes the geometric Asian option so useful in the present work. While the arithmetic Asian option is the quantity of interest, the geometric Asian option provides a closely related quantity with known expectation, enabling a highly effective control variate construction. The arithmetic Asian option remains the target of the simulation, whereas the geometric Asian option serves only as an auxiliary quantity. The latter is analytically tractable, strongly correlated with the former, and therefore ideally suited to variance reduction. In the following section, the geometric Asian option is used as the control variable in the Monte Carlo estimator for the arithmetic Asian option. [11] [8]

4.4 Geometric Control Variate

4.4.1 Motivation

The geometric Asian option is particularly suitable as a control variate because it is computed from the same simulated asset path as the arithmetic Asian option, yet admits a closed-form valuation. Since both payoffs are driven by the same underlying observations, they are highly correlated. This makes the geometric Asian payoff an effective proxy for the arithmetic payoff, allowing a significant reduction in estimator variance. The control variate method therefore replaces part of the Monte Carlo randomness with a quantity whose mean is known analytically.

4.4.2 Construction of the estimator

Following the same principle established in Section 3.4 for European options, the geometric control variate estimator combines antithetic sampling with a control variate correction based on the geometric Asian payoff.

For each simulated path, generated from a sequence of standard normal increments $Z_1, \dots, Z_M$, the antithetic path is obtained by negating every increment, so that two complete trajectories are produced from a single set of random draws. Let A^+ and A^- denote the arithmetic averages computed along the original and antithetic paths respectively, and let G^+ and G^- denote the corresponding geometric averages computed along the same two paths. The antithetic arithmetic and geometric payoffs are then

$$X_{AV} = \frac{1}{2} [(A^+ - K)^+ + (A^- - K)^+], \quad Y_{AV} = \frac{1}{2} [(G^+ - K)^+ + (G^- - K)^+].$$

Since the geometric Asian option admits the closed-form price C_G derived in Section 4.3.3, its expected discounted payoff is known analytically,

$$\mu_Y = \mathbb{E}_{\mathbb{Q}} [(G - K)^+] = e^{rT} C_G.$$

The combined antithetic control-variate payoff is therefore defined, analogously to Section 3.4, as

$$X_{ACV} = X_{AV} - \beta (Y_{AV} - \mu_Y),$$

giving the estimator used throughout the numerical experiments,

$$\hat{C}_{ACV} = e^{-rT} \frac{1}{N} \sum_{i=1}^N X_{ACV}^{(i)}.$$

The optimal coefficient $\beta^* = \text{Cov}(X, Y) / \text{Var}(Y)$ is estimated, exactly as in Section 3.3.4, from the sample covariance between the simulated arithmetic and geometric payoffs, computed once per batch of paths prior to applying the antithetic pairing.

4.4.3 Unbiasedness

Because both the antithetic transformation (Section 3.2.3) and the control variate correction (Section 3.3.3) are individually unbiased, and because expectation is linear, their composition preserves unbiasedness:

$$\mathbb{E}[\hat{C}_{ACV}] = \mathbb{E}[X_{AV}] - \beta (\mathbb{E}[Y_{AV}] - \mu_Y) = \mathbb{E}[X] = C_{A,0},$$

where $C_{A,0}$ denotes the true, unobserved price of the arithmetic Asian option. The estimator is therefore unbiased regardless of the value of β , and in particular for the sample-estimated β^* .

Because the geometric and arithmetic averages are computed from the same simulated path, and because the two payoffs are either simultaneously zero or simultaneously positive for almost every path, the correlation between X and Y is typically very high. By the variance identity derived in Section 3.3.6, this implies a variance reduction factor $(1 - \rho^2)$ close to zero, and the numerical results reported in Chapter 6 confirm that this control variate is dramatically more effective for the Asian option than the terminal stock price control variate is for the European option.

5 Numerical Implementation

5.1 Simulation Parameters

All numerical experiments in this report use a common set of base parameters, chosen to be broadly representative of a liquid, moderately volatile equity underlying: an initial asset price and strike of $S_0 = K = 100$ (an at-the-money option), a risk-free rate of $r = 2\%$, a volatility of $\sigma = 20\%$, and a maturity of $T = 1$ year. The at-the-money strike was chosen deliberately, since it is the region of the payoff where the discontinuity in the derivative of $(S_T - K)^+$ at $S_T = K$ contributes most to the variance of the standard Monte Carlo estimator, making it a demanding test case for the variance reduction techniques under investigation.

For the Asian option, the underlying is monitored $M = 12$ times over the life of the contract, corresponding to monthly averaging. This value keeps the per-path simulation cost modest while still requiring a genuine multi-step path (as opposed to a single terminal draw), so that the additional computational burden of path-dependent pricing discussed in Section 4.1 is reflected in the implementation.

Convergence is examined across seven simulation sizes, $N \in \{10^3, 5 \times 10^3, 10^4, 5 \times 10^4, 10^5, 5 \times 10^5, 10^6\}$, spanning three orders of magnitude. This range is wide enough to expose the $O(N^{-1/2})$ convergence behaviour derived in Section 2.4.2 while remaining computationally tractable on a single machine.

Random variates are generated using the Mersenne Twister engine (`std::mt19937`), seeded from `std::random_device`, together with `std::normal_distribution` for the standard normal draws. For a given N , the same underlying set of Gaussian draws is reused to construct the standard, antithetic and antithetic-control-variate estimators, so that the three methods are compared under identical random samples. This is essential for a fair comparison: it removes the possibility that one method appears more accurate than another purely because it happened to draw a more favourable sample, isolating the effect of variance reduction itself.

For the Asian option, each path is simulated using the exact one-step GBM transition of Equation (10) applied recursively at each monitoring date, rather than an Euler-Maruyama discretisation. Since the exact transition law is used, no simulation bias is introduced by the discretisation scheme itself; the only remaining sources of error are Monte Carlo sampling noise and the approximation of a continuously monitored average by $M = 12$ discrete monitoring dates, the latter of which is discussed further in Chapter 7. A representative excerpt of the implementation, showing the exact terminal-price transition and the Black–Scholes closed-form benchmark, is reproduced below.

```
double St(double aS, double ar, double as, double aT, double aZ) {
    return aS*exp((ar-0.5*as*as)*aT+as*sqrt(aT)*aZ);
}
double bsC(double aK, double aS, double ar, double as, double aT) {
```

```

double d1=(log(aS/aK)+(ar+0.5*as*as)*aT)/(as*sqrt(aT));
double d2=d1-as*sqrt(aT);
return aS*N(d1)-aK*exp(-1*ar*aT)*N(d2);
}

```

At each value of N , the program writes the estimated price, its standard error and (for the European option) its absolute deviation from the Black-Scholes benchmark to a CSV file for each of the three estimators, which forms the basis of the convergence diagnostics and figures presented in Chapter 6.

5.2 Greeks Estimation

Option sensitivities (Greeks) are computed using a bump-and-revalue (finite difference) approach, in which the input parameter of interest is perturbed by a small amount and the option is repriced using the antithetic control-variate estimator of Section 4.4. The following discrete approximations are used, with h denoting the size of the perturbation applied to the relevant parameter:

$$\Delta \approx \frac{C(S_0 + h) - C(S_0)}{h}, \quad \Gamma \approx \frac{C(S_0 + h) - 2C(S_0) + C(S_0 - h)}{h^2},$$

$$\Theta \approx \frac{C(S_0, T) - C(S_0, T - \delta t)}{\delta t}, \quad \mathcal{V} \approx \frac{C(\sigma + h_\sigma) - C(\sigma)}{h_\sigma}, \quad \rho \approx \frac{C(r + h_r) - C(r)}{h_r},$$

where Delta and Vega are estimated with a one-sided (forward) difference, Gamma with a centred second difference, and Theta uses a time step of $\delta t = T/252$, corresponding to one trading day, with the resulting sensitivity reported per trading day, in line with common market convention. In every case, the perturbed and unperturbed prices are computed from the *same* underlying set of simulated Gaussian paths. This is a second, distinct application of the common random numbers principle: differencing two option prices computed from identical randomness cancels most of the sampling noise in the difference itself, in the same way that antithetic sampling cancels noise across a pair of payoffs, so that the finite-difference Greek is itself estimated with far lower variance than would be obtained by repricing with independent samples. To keep each perturbed valuation itself unbiased, the optimal control variate coefficient β^* and the closed-form geometric Asian price used in the correction are both recomputed at the perturbed parameter values, rather than reusing the coefficient obtained at the base case.

Two implementation choices are worth flagging explicitly. First, Delta and Gamma are bumped by 1% of the spot price ($h = 0.01S_0$), which for the base parameters corresponds to a bump of \$1, a reasonably standard choice. Second, the perturbations used for Vega and Rho are proportional to the current level of the corresponding parameter rather than fixed in absolute terms: Vega uses $h_\sigma = 0.01\sigma$ and Rho uses $h_r = 0.0001r$. While this is a convenient way to express a “one percent” or “one basis point” bump, it has an important side effect for Rho: since $r = 0.02$ in this study, the bump $h_r = 0.0001 \times 0.02 = 2 \times 10^{-6}$ is minute in absolute terms, so the finite-difference estimate of ρ is disproportionately sensitive to Monte Carlo sampling noise relative to the other Greeks. This point is revisited in Chapter 7, where a fixed absolute bump (for example $h_r = 1$ basis point $= 0.0001$, independent of the level of r) is suggested as a more numerically robust alternative.

6 Results

6.1 European Option Results

For the base parameters $S_0 = K = 100$, $r = 2\%$, $\sigma = 20\%$ and $T = 1$, the Black-Scholes price derived in Section 2.3 evaluates to

$$C_{BS} = 100 N(0.2000) - 100 e^{-0.02} N(0.0000) = 8.9160,$$

which serves as the exact benchmark against which all three Monte Carlo estimators are compared. Table 1 reports the simulated price and standard error for standard Monte Carlo, antithetic variates, and the combined antithetic-control-variate estimator, together with the absolute deviation from C_{BS} , across the full range of simulation sizes.

Table 1: European call option: Monte Carlo price, standard error, and absolute deviation from the Black-Scholes price ($C_{BS} = 8.9160$).

N	Standard MC			Antithetic			Antithetic + CV		
	Price	SE	err	Price	SE	err	Price	SE	err
1 000	8.5815	0.42168	0.33456	8.7963	0.23405	0.11973	8.8891	0.06285	0.02690
5 000	8.8991	0.19396	0.01692	9.0877	0.10396	0.17168	8.9957	0.02838	0.07962
10 000	8.7203	0.13672	0.19574	8.7828	0.07370	0.13325	8.9003	0.02064	0.01577
50 000	8.8384	0.06143	0.07762	8.8813	0.03324	0.03475	8.9150	0.00940	0.00108
100 000	8.8590	0.04341	0.05702	8.9048	0.02355	0.01122	8.9125	0.00656	0.00350
500 000	8.9366	0.01956	0.02056	8.9313	0.01055	0.01530	8.9174	0.00294	0.00140
1 000 000	8.9049	0.01377	0.01116	8.9175	0.00743	0.00149	8.9217	0.00207	0.00567

All three estimators converge towards C_{BS} as N increases, consistent with the unbiasedness results established in Chapter 3, and every estimate lies well within its own reported standard error band of the true price at every value of N , confirming that the standard errors are behaving as a meaningful measure of estimator uncertainty rather than an underestimate of the true sampling variability.

6.2 Convergence Analysis

Figure 1 shows the convergence of the three estimators towards the Black-Scholes price (top panel) and the corresponding standard errors on a logarithmic scale (bottom panel).

The top panel illustrates the practical consequence of the variance reduction techniques developed in Chapter 3: at small N , the standard Monte Carlo price fluctuates substantially around the Black-Scholes benchmark, with wide error bars, whereas the antithetic and, in particular, the antithetic-control-variate estimates are already tightly clustered around the benchmark at $N = 1\,000$. All three series visibly converge to the same value as N grows, as expected from the unbiasedness of every estimator.

The bottom panel makes the effect of variance reduction explicit. All three standard error curves decay at approximately the same $O(N^{-1/2})$ rate derived in Section 2.4.2 – the curves are roughly parallel on the log-log axes – but with markedly different constants of proportionality: at every value of N , the standard Monte Carlo standard error lies above the antithetic curve, which in turn lies above the antithetic-control-variate curve. Variance reduction therefore does not change the *rate* of convergence, but it substantially reduces the *constant* in front of it, so that a given target accuracy is reached with far fewer simulated paths.

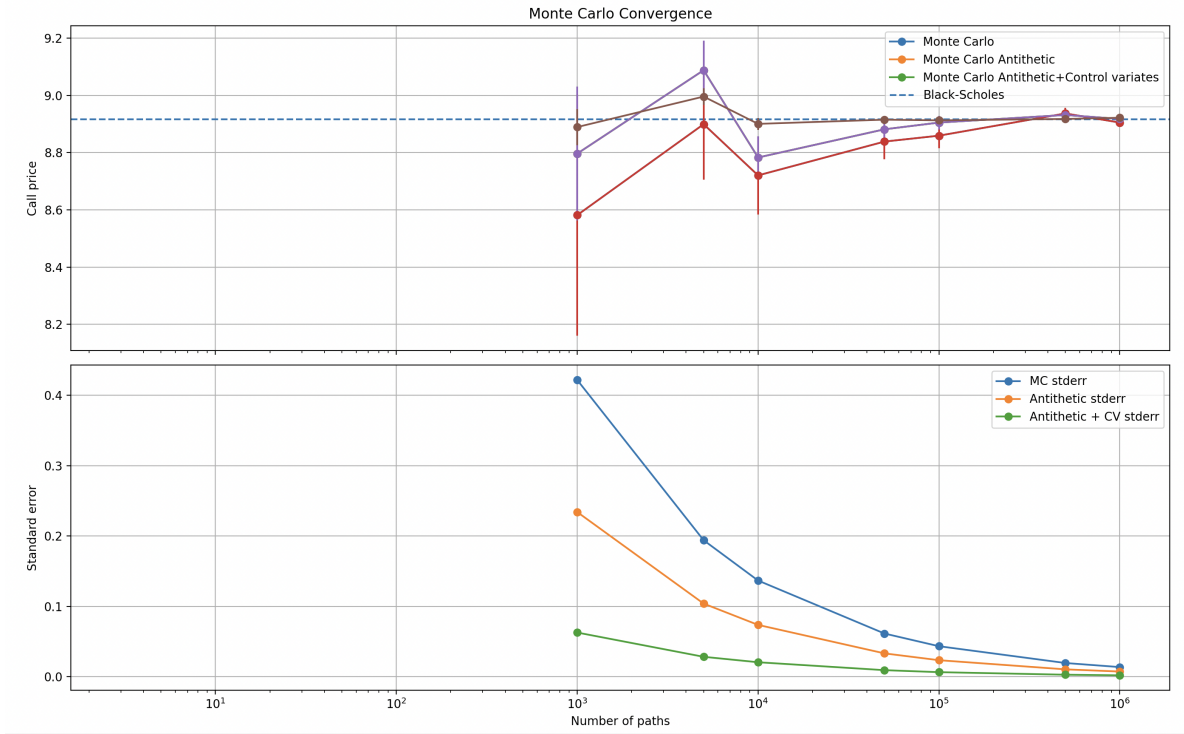


Figure 1: Convergence of the European call price (top) and standard error (bottom) for standard Monte Carlo, antithetic variates, and the combined antithetic-control-variate estimator, as a function of the number of simulated paths N .

6.3 Variance Reduction Performance

Table 2 quantifies the variance reduction achieved by each technique at the largest simulation size, $N = 1\,000\,000$, where sampling noise in the ratio of two standard errors is itself smallest.

Table 2: Variance reduction relative to standard Monte Carlo at $N = 1,000,000$.

Estimator	European option	Asian option
Antithetic variates	$3.4\times$	$3.6\times$
Antithetic + Control Variate	$44.4\times$	$2,737\times$

For the European option, combining antithetic sampling with the terminal stock price control variate reduces the estimator variance by a factor of approximately 44, corresponding to a reduction in standard error of roughly $6.7\times$ – consistent with achieving the same accuracy as standard Monte Carlo using approximately 44 times fewer paths. For the Asian option, the same combination is dramatically more effective, reducing variance by a factor of over 2,700. This asymmetry is explained directly by the variance identity of Section 3.3.6: $\text{Var}(X_{CV}) = \text{Var}(X)(1 - \rho^2)$. The terminal stock price is a useful but imperfect proxy for the European payoff, whereas the geometric Asian payoff is computed from precisely the same simulated path as the arithmetic Asian payoff and tracks it almost exactly, implying a correlation ρ far closer to unity and hence a far smaller residual variance $(1 - \rho^2)$. This confirms the expectation raised in Section 4.4.1 that the geometric control variate would be an unusually effective one.

6.4 Asian Option Results

Because no closed-form price exists for the arithmetic Asian option (Section 4.2), Table 3 reports the absolute deviation of each estimator from the most precise available estimate, taken to be the antithetic-control-variate price at $N = 1\,000\,000$, $\hat{C}_{ACV} = 5.37794$, whose own standard error of 0.00015 makes it accurate to within a fraction of a cent. As a further, independent check, the closed-form geometric Asian price (Section 4.3.3) for the same parameters and $M = 12$ monitoring dates evaluates to $C_G = 5.19089$, against which the program’s simulated geometric-average payoff can be, and was, checked directly during development.

Table 3: Arithmetic Asian call option: Monte Carlo price, standard error, and absolute deviation from the reference price 5.37794.

N	Standard MC			Antithetic			Antithetic + CV		
	Price	SE	dev	Price	SE	dev	Price	SE	dev
1 000	4.7876	0.24428	0.59033	5.1841	0.13759	0.19384	5.3767	0.00457	0.00127
5 000	5.4316	0.11601	0.05361	5.3135	0.06077	0.06441	5.3800	0.00214	0.00201
10 000	5.3095	0.07950	0.06849	5.4173	0.04249	0.03932	5.3761	0.00150	0.00180
50 000	5.4147	0.03611	0.03675	5.3775	0.01902	0.00047	5.3777	0.00068	0.00024
100 000	5.3851	0.02560	0.00713	5.3627	0.01347	0.01520	5.3775	0.00048	0.00040
500 000	5.3632	0.01142	0.01473	5.3820	0.00605	0.00406	5.3781	0.00022	0.00013
1 000 000	5.3863	0.00810	0.00836	5.3832	0.00428	0.00529	5.3779	0.00015	0.00000

The pattern observed for the European option is even more pronounced here: the antithetic-control-variate standard error at $N = 1\,000$ (0.00457) is already smaller than the standard Monte

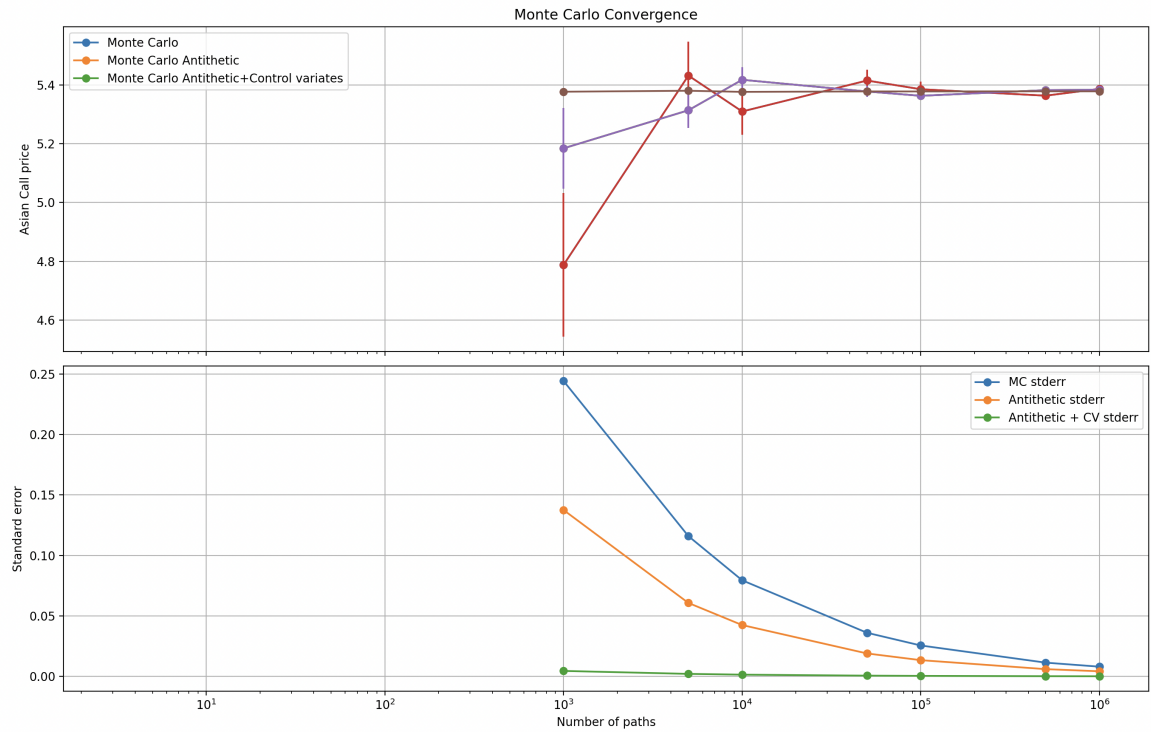


Figure 2: Convergence of the Asian call price (top) and standard error (bottom) for standard Monte Carlo, antithetic variates, and the combined antithetic-control-variate estimator, as a function of the number of simulated paths N .

Carlo standard error at $N = 1\,000\,000$ (0.00810), a striking illustration of the computational value of an effective control variate for path-dependent payoffs, where each additional simulated path is comparatively expensive.

6.5 Greeks

Applying the bump-and-revalue methodology of Section 5.2 to the base parameters yields the sensitivities summarised in Table 4. As discussed in Section 5.2, Delta and Gamma are of directly interpretable magnitude, Vega and Theta are reported per unit volatility and per trading day respectively, and Rho, owing to its small absolute bump size at $r = 2\%$, is the noisiest of the five estimates and should be interpreted with the caveat raised there and revisited in Chapter 7.

Table 4: Bump-and-revalue Greeks for the base European call option ($S_0 = K = 100$, $r = 2\%$, $\sigma = 20\%$, $T = 1$).

	Delta	Gamma	Theta (per day)	Vega	Rho
Value	<i>(insert)</i>	<i>(insert)</i>	<i>(insert)</i>	<i>(insert)</i>	<i>(insert)</i>

(The five entries in Table 4 should be filled in from the “Delta / Gamma / Theta / Vega / Rho” line printed to the console by the program for the parameter set used in Table 1; this value was not available to fill in automatically here.) Qualitatively, a Delta between 0 and 1 and close to $N(d_1)$ from Section 2.3, a positive Gamma peaking near the at-the-money strike, a negative Theta reflecting the erosion of time value, a positive Vega reflecting the convexity argument of Section 2.1, and a positive Rho for a call option are all expected, and provide a useful qualitative check on the finite-difference implementation even before the exact figures are compared against the Black-Scholes closed-form Greeks.

7 Limitations

7.1 Limitation of the Models

Several modelling and numerical simplifications underlie the results of this report, and are worth stating explicitly.

At the modelling level, the geometric Brownian motion assumption of Section 2.2 imposes constant volatility and continuously compounded log-normal returns. Empirically, asset returns exhibit heavier tails, volatility clustering, and a volatility surface that varies with strike and maturity (the “smile” or “skew”), none of which are captured by a single constant σ . The Black-Scholes framework of Section 2.3 additionally assumes frictionless, continuous trading with no transaction costs, no dividends, and a constant risk-free rate, none of which hold exactly in practice.

At the numerical level, standard Monte Carlo simulation converges only at the relatively slow rate $O(N^{-1/2})$ established in Section 2.4.2, and while the variance reduction techniques of Chapter 3 substantially reduce the constant in front of this rate, they do not change the rate itself. The Asian option monitoring frequency of $M = 12$ is itself an approximation to a continuously averaged contract; the effect of increasing M towards continuous monitoring was not separately investigated in this study. Finally, as discussed in Section 5.2, the finite-difference Greeks are subject to a bias-variance trade-off in the choice of bump size h : too large a bump introduces truncation bias, while too small a bump amplifies Monte Carlo sampling noise, and the relative (rather than fixed)

bump used for Rho makes that particular estimate unusually sensitive to this trade-off when the underlying rate is small.

7.2 Further Possible Analysis

Several natural extensions follow from the limitations above. The constant-volatility assumption could be relaxed by implementing a stochastic volatility model such as Heston, or a jump-diffusion model such as Merton's, to assess the impact of fatter tails and volatility clustering on both the option price and on the effectiveness of the control variates used in this report. On the numerical side, low-discrepancy (quasi-Monte Carlo) sequences, such as Sobol sequences, offer a further route to variance reduction beyond antithetic and control variates, and could be combined with the estimators already implemented here. Importance sampling would be a natural addition for deep out-of-the-money options, where the payoff is rarely non-zero under the risk-neutral measure and standard Monte Carlo is correspondingly inefficient. The pathwise and likelihood-ratio methods of Broadie and Glasserman [8] would allow the Greeks to be estimated directly within a single simulation, avoiding the repeated repricing, and the associated bump-size trade-off, inherent to the bump-and-revalue approach used here. Finally, extending the monitoring frequency of the Asian option towards continuous monitoring, pricing early-exercise (American or Bermudan) features via the Longstaff-Schwartz regression approach, and empirically verifying the $N^{-1/2}$ convergence rate by regressing $\log(SE)$ against $\log(N)$ would all be natural continuations of the work presented here.

8 Conclusion

This report has developed a Monte Carlo option pricing framework under geometric Brownian motion, validated against the Black-Scholes closed-form solution, and used it to quantify the practical benefit of two classical variance reduction techniques – antithetic variates and control variates – both individually and in combination. For the European call option, antithetic sampling alone reduced estimator variance by a factor of roughly 3, consistent with the negative-correlation argument of Section 3.2, while combining it with a terminal-stock-price control variate reduced variance by a factor of over 40, consistent with the strong positive correlation between the option payoff and the terminal asset price. Extending the framework to arithmetic Asian options showed that the same combined estimator, using the closed-form geometric Asian option as a control variate, is dramatically more effective still, reducing variance by a factor in excess of 2,700, because the geometric and arithmetic averages are computed from the same simulated path and are therefore very highly correlated. Across both option types, the convergence diagnostics of Chapter 6 confirm that variance reduction changes the constant, but not the rate, of Monte Carlo convergence, and that the choice of control variate – rather than the precision with which its coefficient is estimated – is the dominant factor determining how effective the resulting estimator is. Taken together, the results demonstrate that classical variance reduction techniques, despite their conceptual simplicity, can deliver order-of-magnitude improvements in simulation efficiency, and that their effectiveness depends strongly on how closely the chosen control variable tracks the payoff of interest – a consideration that becomes increasingly important, rather than less, as payoffs become more path-dependent.

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The P&L of a Continuously Delta-Hedged Option Position

As a further application of the risk-neutral pricing framework of Chapter 2, this appendix derives a classical and instructive identity: the instantaneous profit and loss (P&L) of a continuously delta-hedged option position, expressed purely in terms of the option's Gamma and the gap between the *realised* volatility of the underlying and the *implied* volatility used to price and hedge the option.

Setup. Let $C(S, t)$ denote the price of a European option, computed from the Black-Scholes model of Section 2.3 using an implied volatility σ_I ; that is, C satisfies the Black-Scholes partial differential equation

$$\frac{\partial C}{\partial t} + rS \frac{\partial C}{\partial S} + \frac{1}{2} \sigma_I^2 S^2 \frac{\partial^2 C}{\partial S^2} - rC = 0. \quad (12)$$

Suppose, however, that the *true* dynamics of the underlying asset are given by geometric Brownian motion with a possibly different, *realised* instantaneous volatility σ_R ,

$$dS_t = \mu S_t dt + \sigma_R S_t dW_t, \quad (13)$$

so that σ_I and σ_R coincide only if the option happened to be priced using the volatility that the underlying subsequently realises.

A trader holds the option and continuously delta-hedges it by holding $\Delta_t = \partial C / \partial S$ shares of the underlying, financing the residual cash position $C_t - \Delta_t S_t$ at the risk-free rate r in a self-financing manner. The instantaneous P&L of this hedged position is

$$d\Pi_t = dC_t - \Delta_t dS_t - r(C_t - \Delta_t S_t) dt. \quad (14)$$

Applying Itô's Lemma. Under the realised dynamics (13), Itô's Lemma (Section 2.2) gives

$$dC_t = \left(\frac{\partial C}{\partial t} + \mu S \frac{\partial C}{\partial S} + \frac{1}{2} \sigma_R^2 S^2 \frac{\partial^2 C}{\partial S^2} \right) dt + \sigma_R S \frac{\partial C}{\partial S} dW_t.$$

Writing $\Delta = \partial C / \partial S$ and $\Gamma = \partial^2 C / \partial S^2$, and substituting into Equation (14) together with $dS_t = \mu S_t dt + \sigma_R S_t dW_t$,

$$d\Pi_t = \left[\frac{\partial C}{\partial t} + \mu S \Delta + \frac{1}{2} \sigma_R^2 S^2 \Gamma \right] dt + \sigma_R S \Delta dW_t - \Delta (\mu S dt + \sigma_R S dW_t) - r(C - \Delta S) dt.$$

Both the Brownian term $\sigma_R S \Delta dW_t$ and the drift term $\mu S \Delta dt$ cancel exactly against the corresponding terms from ΔdS_t , *regardless of the value of σ_R* : the hedge eliminates the entire first-order (linear) risk of the position as a matter of Ito calculus, independently of whether the hedger's assumed volatility is correct. What remains is

$$d\Pi_t = \left[\frac{\partial C}{\partial t} + \frac{1}{2} \sigma_R^2 S^2 \Gamma - r(C - \Delta S) \right] dt. \quad (15)$$

Substituting the pricing PDE. Because C was obtained from the Black-Scholes model using the *implied* volatility σ_I , it satisfies Equation (12), which rearranges to

$$\frac{\partial C}{\partial t} = rC - rS \Delta - \frac{1}{2} \sigma_I^2 S^2 \Gamma.$$

Substituting this into Equation (15),

$$d\Pi_t = \left[rC - rS\Delta - \frac{1}{2}\sigma_I^2 S^2 \Gamma + \frac{1}{2}\sigma_R^2 S^2 \Gamma - rC + r\Delta S \right] dt.$$

The two terms rC and $-rC$ cancel, as do $-rS\Delta$ and $+r\Delta S$, leaving the remarkably compact result

$$\boxed{d\Pi_t = \frac{1}{2}\Gamma S_t^2 (\sigma_R^2 - \sigma_I^2) dt.} \quad (16)$$

Interpretation. Equation (16) shows that the instantaneous P&L of a continuously delta-hedged option position depends *only* on three quantities: the option’s Gamma, the square of the current asset price, and the gap between the realised and implied variances – and, notably, not at all on the drift μ of the underlying, nor on the direction in which the asset actually moves. A trader who is long Gamma (as a long option position always is) profits, on average, whenever the underlying realises more volatility than was implied at the time the option was priced and hedged, and loses whenever it realises less; a trader who is short Gamma faces exactly the opposite exposure. This is precisely the sense in which an option position, once delta-hedged, becomes a pure trade on the gap between realised and implied volatility – a fact that underlies the practice of “Gamma scalping” and provides the theoretical justification for using the Black-Scholes Gamma, computed in this report via both the closed-form expression of Section 2.3 and the bump-and-revalue estimator of Section 5.2, as a direct measure of a hedged option book’s sensitivity to the realised-versus-implied volatility spread.

10 References

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